



Mathematics:

Sequence And Progressions

CHAPTER AT A GLANCE: CORE FORMULAS & RULES

- **Sequence:** An ordered list of numbers where each number is a *term* ($t_1, t_2, \dots, t_n$).
- **Explicit Rule:** Calculates term value using position n directly (e.g., $t_n = 2n - 1$).
- **Recursive Rule:** Expresses a term using preceding term(s) (e.g., $t_n = t_{n-1} + d$).
- **Virahāṅka–Fibonacci Sequence:** $V_1 = 1, V_2 = 2, V_n = V_{n-1} + V_{n-2}$ for $n \geq 3$ (studied in 7th c. CE).
- **Arithmetic Progression (AP):** Constant common difference $d = t_n - t_{n-1}$.
 - n^{th} term: $t_n = a + (n - 1)d$ (Linear graph)
 - Sum of first n natural numbers: $S_n = \frac{n(n+1)}{2}$ (Āryabhata, c. 499 CE)
- **Geometric Progression (GP):** Constant common ratio $r = \frac{t_n}{t_{n-1}}$.
 - n^{th} term: $t_n = a \cdot r^{n-1}$ (Exponential growth/decay graph)

. Multiple Choice Questions (1 Mark Each)

Q1. Sequence & Rule Identification

Which of the following sequences follows an explicit rule given by $u_n = 5n - 2$?

- (A) 3, 8, 13, 18, ...
- (B) 3, 7, 11, 15, ...
- (C) 5, 10, 15, 20, ...
- (D) 2, 5, 8, 11, ...

Q2. Virahāṅka–Fibonacci Sequence

In the Virahāṅka–Fibonacci sequence with $V_1 = 1$ and $V_2 = 2$, what is the 6th term (V_6)?

- (A) 8
- (B) 13
- (C) 21
- (D) 34

Q3. Common Difference in AP

What is the common difference (d) of the Arithmetic Progression: 11, 7, 3, -1, -5, ...?

- (A) 4
- (B) -4
- (C) 3
- (D) -3

Q4. Geometric Progression Property

When plotting ordered pairs (n, t_n) emerging from a Geometric Progression (GP) on a graph:

- (A) They always lie on a straight line.
- (B) They do not lie on a straight line, forming a rapid rising or falling curve.
- (C) They form a vertical straight line.
- (D) They form a horizontal line parallel to the x-axis.

Q5. Sum of Natural Numbers

The sum of the first 20 natural numbers $(1 + 2 + 3 + \dots + 20)$ is:

- (A) 190
- (B) 200
- (C) 210
- (D) 220

Q6. Assertion & Reason

Assertion (A): The sequence $2, 2\sqrt{2}, 4, 8, \dots$ is an Arithmetic Progression.

Reason (R): An Arithmetic Progression requires a constant common difference between consecutive terms.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
- (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
- (C) (A) is false but (R) is true.
- (D) Both (A) and (R) are false.

I. Short Answer Questions (2–3 Marks Each)**Q7. Testing Term Belongingness**

Consider the explicit formula $t_n = 3n - 7$ for $n \geq 1$.

- (i) Calculate the 1st, 3rd, and 18th terms.
- (ii) Determine whether 332 is a term of this sequence. If so, find its position n .
- (iii) Is 557 a term of this sequence? Explain why or why not.

Q8. Recursive Sequence Evaluation

A sequence is defined by the recursive rule $u_1 = 1$, and $u_n = 2u_{n-1} + 3$ for $n \geq 2$. Calculate the first four terms (u_1, u_2, u_3, u_4) of this sequence.

Q9. AP Term and Position Calculation

Find the 10th and 26th terms of the AP: $3, 8, 13, 18, \dots$. Which term of this AP is 178?

Q10. Range Sum of Natural Numbers

Calculate the sum of consecutive numbers $25 + 26 + 27 + \cdots + 58$ using the formula for the sum of the first n natural numbers (S_n).

Q11. Bouncing Ball GP Application

A ball is dropped from a height of 24 feet. After each bounce, it reaches $\frac{3}{4}$ of its previous height.

- (i) Write the sequence of maximum heights reached after the first 3 bounces.
- (ii) State the explicit rule for the height after the n^{th} bounce.

Q12. Real-World Salary Progression

Harish started work at an annual salary of Rs. 5,00,000 and received an annual increment of Rs. 20,000 each year. After how many years will his annual income reach Rs. 7,00,000?

II. Long Answer & Proof Questions (4–5 Marks Each)

Q13. Determining an AP from System of Equations

Find the 31st term of an Arithmetic Progression whose 11th term is 38 and whose 16th term is 73.

Q14. Multiples in Range

How many three-digit numbers are divisible by 7? Show all steps clearly by setting up the appropriate Arithmetic Progression.

Q15. Geometric Progression with Product Condition

The sum of the first three terms of a GP is $\frac{13}{12}$ and their product is -1 . Find the common ratio r and the three terms of the GP.

Q16. Algebraic Proof in GP

If the 4th, 10th, and 16th terms of a GP are x , y , and z respectively, prove that x, y, z are themselves in Geometric Progression ($y^2 = xz$).

Q17. Fractal Application: Sierpiński Carpet

Stage 0 of the Sierpiński square carpet is a solid square sheet of area 1 sq unit. In Stage 1, it is divided into 9 equal squares and the central square is removed. This process is repeated indefinitely.

- (i) Write the number of red squares remaining at Stages 0, 1, 2, 3, 4.
- (ii) State the explicit and recursive rules for the number of red squares at Stage n .
- (iii) State the explicit rule for the area of the red region at Stage n , and explain what happens to this area as $n \rightarrow \infty$.

III. Complete Solutions & Answer Key

Solutions: Section I (Multiple Choice Questions)

Q1 Solution: (A)

Substitute $n = 1, 2, 3$ into $u_n = 5n - 2$:

- $u_1 = 5(1) - 2 = 3$
- $u_2 = 5(2) - 2 = 8$
- $u_3 = 5(3) - 2 = 13$

Thus, the sequence is 3, 8, 13, 18, ..., which matches Option (A).

Q2 Solution: (B)

The Virahāṅka–Fibonacci rule is $V_n = V_{n-1} + V_{n-2}$ with $V_1 = 1, V_2 = 2$:

- $V_3 = V_2 + V_1 = 2 + 1 = 3$
- $V_4 = V_3 + V_2 = 3 + 2 = 5$
- $V_5 = V_4 + V_3 = 5 + 3 = 8$
- $V_6 = V_5 + V_4 = 8 + 5 = 13$

Hence, $V_6 = 13$ (Option B).

Q3 Solution: (B)

$d = t_2 - t_1 = 7 - 11 = -4$. Check $t_3 - t_2 = 3 - 7 = -4$. Common difference $d = -4$ (Option B).

Q4 Solution: (B)

Points emerging from an AP lie on a straight line. Points from a GP do **not** lie on a straight line because terms grow/decay exponentially (Option B).

Q5 Solution: (C)

Using Āryabhata's formula $S_n = \frac{n(n+1)}{2}$ for $n = 20$:

$$S_{20} = \frac{20 \times 21}{2} = 10 \times 21 = 210 \quad (\text{Option C})$$

Q6 Solution: (C)

Checking consecutive differences: $2\sqrt{2} - 2 \approx 0.828$, while $4 - 2\sqrt{2} \approx 1.172$. Differences are not constant, so it is a GP ($r = \sqrt{2}$), not an AP. Assertion (A) is false. Reason (R) is a correct mathematical definition of AP. Hence, (C) is correct.

Solutions: Section II (Short Answer Questions)

Q7 Solution:

Given explicit rule: $t_n = 3n - 7$.

- (i) $t_1 = 3(1) - 7 = -4$; $t_3 = 3(3) - 7 = 2$; $t_{18} = 3(18) - 7 = 54 - 7 = 47$.
- (ii) Set $3n - 7 = 332 \implies 3n = 339 \implies n = 113$. Since 113 is a natural number, 332 is the **113th term**.
- (iii) Set $3n - 7 = 557 \implies 3n = 564 \implies n = 188$. Since 188 is a natural number, 557 is the **188th term**.

Q8 Solution:

Given $u_1 = 1$, $u_n = 2u_{n-1} + 3$:

- $u_1 = 1$
- $u_2 = 2(1) + 3 = 5$
- $u_3 = 2(5) + 3 = 13$
- $u_4 = 2(13) + 3 = 29$

First four terms are 1, 5, 13, 29.

Q9 Solution:

For AP 3, 8, 13, 18, ...: First term $a = 3$, common difference $d = 8 - 3 = 5$.

- $t_{10} = a + 9d = 3 + 9(5) = 3 + 45 = \mathbf{48}$
- $t_{26} = a + 25d = 3 + 25(5) = 3 + 125 = \mathbf{128}$
- To find position of 178: $3 + (n-1)5 = 178 \implies 5(n-1) = 175 \implies n-1 = 35 \implies \mathbf{n = 36}$.

Q10 Solution:

We can write $25 + 26 + \dots + 58 = S_{58} - S_{24}$.

$$S_{58} = \frac{58 \times 59}{2} = 29 \times 59 = 1711$$

$$S_{24} = \frac{24 \times 25}{2} = 12 \times 25 = 300$$

$$\text{Sum} = 1711 - 300 = \mathbf{1411}$$

Q11 Solution:

Initial height = 24 ft, bounce ratio $r = 0.75$.

- (i) $t_1 = 24 \times 0.75 = \mathbf{18}$ ft, $t_2 = 18 \times 0.75 = \mathbf{13.5}$ ft, $t_3 = 13.5 \times 0.75 = \mathbf{10.125}$ ft.
- (ii) Explicit formula: $\mathbf{t_n = 18 \times (0.75)^{n-1}}$.

Q12 Solution:

Income forms an AP with $a = 5,00,000$ and $d = 20,000$.

$$t_n = a + (n-1)d = 7,00,000$$

$$5,00,000 + (n-1)20,000 = 7,00,000 \implies (n-1)20,000 = 2,00,000$$

$$n - 1 = 10 \implies n = 11$$

Since $n = 11$ represents the 11th year, the income reaches Rs. 7,00,000 after 10 **years** of completed service.

Solutions: Section III (Long Answer Questions)

Q13 Solution:

Let the first term be a and common difference be d .

$$t_{11} = a + 10d = 38 \quad \text{--- (Equation 1)}$$

$$t_{16} = a + 15d = 73 \quad \text{--- (Equation 2)}$$

Subtracting Eq. 1 from Eq. 2:

$$5d = 35 \implies d = 7$$

Substitute $d = 7$ into Eq. 1:

$$a + 10(7) = 38 \implies a + 70 = 38 \implies a = -32$$

Calculating the 31st term:

$$t_{31} = a + 30d = -32 + 30(7) = -32 + 210 = 178$$

Q14 Solution:

Three-digit numbers range from 100 to 999.

- Smallest 3-digit multiple of 7: $100 \div 7 = 14.28 \implies 7 \times 15 = 105 = a$.
- Largest 3-digit multiple of 7: $999 \div 7 = 142.71 \implies 7 \times 142 = 994 = t_n$.
- Common difference $d = 7$.

Using $t_n = a + (n - 1)d$:

$$994 = 105 + (n - 1)7 \implies 889 = (n - 1)7 \implies n - 1 = 127 \implies n = 128$$

There are 128 three-digit numbers divisible by 7.

Q15 Solution:

Let the three terms of the GP be $\frac{a}{r}, a, ar$.

Step 1: Product condition: $\left(\frac{a}{r}\right) \times a \times (ar) = -1 \implies a^3 = -1 \implies a = -1$.

Step 2: Sum condition: $\frac{a}{r} + a + ar = \frac{13}{12}$. Substitute $a = -1$:

$$\frac{-1}{r} - 1 - r = \frac{13}{12} \implies -\left(\frac{1}{r} + 1 + r\right) = \frac{13}{12}$$

Multiply by $-12r$: $12 + 12r + 12r^2 = -13r \implies 12r^2 + 25r + 12 = 0$.

Step 3: Factorize quadratic: $12r^2 + 16r + 9r + 12 = 0 \implies (4r + 3)(3r + 4) = 0$. Thus, $r = -\frac{3}{4}$ or $r = -\frac{4}{3}$.

Terms: If $r = -3/4$, terms are $\frac{4}{3}, -1, \frac{3}{4}$. If $r = -4/3$, terms are $\frac{3}{4}, -1, \frac{4}{3}$.

Q16 Solution:

Let GP have first term a and common ratio r .

$$x = t_4 = ar^3, \quad y = t_{10} = ar^9, \quad z = t_{16} = ar^{15}$$

To prove x, y, z are in GP, we verify $y^2 = xz$:

$$\text{LHS} = y^2 = (ar^9)^2 = a^2r^{18}$$

$$\text{RHS} = x \times z = (ar^3) \times (ar^{15}) = a^2r^{3+15} = a^2r^{18}$$

Since $\text{LHS} = \text{RHS} = a^2r^{18}$, $y^2 = xz$. Thus, x, y, z are in Geometric Progression. ■

Q17 Solution:

- (i) Number of red squares: Stage 0 = 1, Stage 1 = 8, Stage 2 = 64 = 8^2 , Stage 3 = 512 = 8^3 , Stage 4 = 4096 = 8^4 .
- (ii) Explicit formula: $\mathbf{t_n = 8^n}$ for $n \geq 0$. Recursive formula: $\mathbf{t_0 = 1, t_n = 8t_{n-1}}$ for $n \geq 1$.
- (iii) At each stage, $\frac{1}{9}$ of the remaining area is removed, keeping $\frac{8}{9}$. Explicit formula for area: $\mathbf{s_n = \left(\frac{8}{9}\right)^n}$. As $n \rightarrow \infty$, since $\frac{8}{9} < 1$, $\left(\frac{8}{9}\right)^n \rightarrow 0$. The shaded area approaches **zero**.